An Event-Level Database of High-Frequency Crypto Surprises[☆]

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Abstract

This document describes the construction of an event-level database of crypto surprises. It combines a narrative identification approach with a high-frequency measurement strategy to isolate unanticipated, cryptocurrency-specific price movements around major news events. Starting from the full corpus of Reuters cryptocurrency news, the analysis identifies candidate events through comprehensive narrative reading, validates each event against intraday Bitcoin price data using both human researchers and a large language model, and constructs a daily series of crypto surprises based on tight intraday return windows. The resulting dataset contains detailed event-level information, including event timestamps, classification outcomes, certainty ratings, and supporting documentation for each candidate event.

[☆]The full event-by-event documentation, including the LLM classification output, is available at `https://ambropo.github.io/CryptoShocks/index.html`. The views expressed in this paper are those of the authors, and not necessarily those of the Bank of England, the Banque de France, the Eurosystem, or their committees.

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Contents

1 Overview **3**

2 Stage 1: Event Identification **5**
2.1 Pillar 1: A Reliable Source 5
2.2 Pillar 2: A Clear Idea of What One Is Looking For 6
2.3 Pillar 3: Approaching the Source Dispassionately and Consistently 7
2.4 Pillar 4: Documenting the Evidence 8
2.5 Final list of events 9

3 Stage 2: Event Validation **9**
3.1 Human Validation 11
3.2 LLM-Assisted Validation 12
3.3 Relationship Between the Two Approaches 13

4 Final Dataset and Variable Definitions **14**

5 Event Classification and Agreement **17**

6 Measuring Crypto Surprises & Stylized Facts **20**
6.1 Definition of the Surprises 20
6.2 Properties of the Surprise Series: All, Human, and LLM 21
6.3 Properties of the Surprise Series: Window Length 23

References **25**

1 Overview

This document describes how we construct the event-level dataset of high-frequency crypto surprises used in the main paper, Cesa-Bianchi and Pinchetti (2026). The central challenge is that, unlike in standard applications of narrative or high-frequency identification, the cryptocurrency sector does not provide a predefined and finite set of candidate events. In settings such as monetary policy, researchers begin from a known sequence of policy announcements and ask which of them contain exogenous shocks. In our setting, by contrast, the researcher must first identify which pieces of news are potentially relevant for the market from a large and continuously evolving information set, and only then determine whether they represent clean exogenous surprises.

We address this problem with a two-stage procedure. In Stage 1, we identify candidate events through a comprehensive narrative reading of the Reuters News Archive. In Stage 2, we validate each candidate event against intraday Bitcoin price data using two independent approaches: a manual validation conducted by human researchers and a structured validation conducted by a large language model (LLM). Stage 2 therefore performs both *validation* and *measurement*: it determines whether the event generates a clean directional market response, and it quantifies that response using high-frequency Bitcoin prices in a narrow intraday window.

Our approach combines two identification strategies that are widely used in macroeconomics. The *narrative approach*, associated with the work of Friedman and Schwartz (1963), Romer and Romer (2004), and Romer and Romer (2010), uses contemporaneous qualitative sources—such as newspapers, government records, or policy documents—to identify episodes that are informative about causal relationships.[1] The value of narrative evidence lies in the fact that it provides information about the *nature* of a shock that is not directly available in quantitative data alone. The *high-frequency approach*, associated with Kuttner (2001) and Gürkaynak et al. (2005), measures asset-price changes in tight windows around events so that the observed movement can be credibly attributed to the news rather than to confounding developments.[2]

We adapt these two approaches to the cryptocurrency sector. Our objective is to construct a series of *crypto surprises*: episodes in which new information specific to the cryptocurrency market plausibly moves prices, but where the underlying news is unrelated to traditional macroeconomic or financial conditions. This distinction is central for the

[1] See Romer and Romer (2023) for a recent synthesis and methodological discussion.

[2] Related applications include Gertler and Karadi (2015) for monetary policy, Känzig (2021) for oil supply news, and Känzig (2023) for carbon pricing.

broader project. While macroeconomic and financial shocks undoubtedly affect cryptocurrency prices, our aim is to isolate shocks that originate *within* the crypto ecosystem, so that the main paper can study how such shocks propagate to traditional financial markets and, more broadly, to the real economy.

A key feature of our setting is the *discovery problem*. In standard applications of narrative and high-frequency identification, the universe of candidate episodes is usually known in advance. For example, every FOMC meeting is a potential carrier of a monetary policy shock. In the cryptocurrency sector, there is no comparable enumeration. Relevant news may concern regulatory actions across jurisdictions, institutional adoption, exchange incidents, technological changes, or public statements by influential actors. The event set is therefore not given ex ante. Stage 1 addresses this discovery problem by identifying candidate events from a reliable contemporaneous narrative source. Stage 2 then disciplines that initial selection by confronting each candidate event with intraday market evidence.

This two-stage design also allows us to introduce a reproducible validation layer into the narrative approach. In particular, we use an LLM as a structured second reader. The role of the LLM is not to discover events or to generate unconstrained interpretations, but to apply a fixed validation protocol to the same candidate events considered by the human researchers. This makes it possible to assess the extent to which narrative classification can be systematized, while preserving the economic discipline of the underlying event-selection exercise.

This methodological appendix focuses exclusively on the construction of the event-level dataset. It should be read together with the other components of the submission package. The main paper uses the validated surprise series in the asset-pricing analysis. The slide deck summarizes the motivation, methodology, and main empirical results. The file `event_instructions.md` reports the detailed instructions used to implement the event-level narrative validation, and the files `event_report.csv` and `event_dataset.html` provide the structured and human-readable outputs of the event-level classification exercise.

The rest of this document proceeds as follows. Section 2 describes Stage 1, namely the identification of candidate events from the Reuters News Archive, organized around the four requirements for rigorous narrative analysis emphasized by Romer and Romer (2023). Section 3 describes Stage 2, namely the validation of candidate events against intraday Bitcoin price data, covering both the human and LLM-assisted procedures. Section 3.3 discusses the relationship between the two validation approaches. Section 4 presents the final dataset and variable definitions. Section **??** documents the classification outcomes, the agreement between validators, and the composition of events by type. Section 6 describes the construction of the surprise series and its main stylized facts.

2 Stage 1: Event Identification

Romer and Romer (2023) identify four requirements for rigorous narrative analysis: (i) a reliable narrative source; (ii) a clear idea of what one is looking for in the source; (iii) approaching the source dispassionately and consistently; and (iv) documenting the narrative evidence carefully. We organize our description of Stage 1 around these four pillars.

2.1 Pillar 1: A Reliable Source

The first requirement for rigorous narrative analysis is a reliable source that is real-time, consistent over time, and detailed and accurate (Romer and Romer, 2023). In their monetary policy application, Romer and Romer use the historical *Minutes* and *Transcripts* of FOMC meetings—documents that are contemporaneous with the policy decisions, exist for the full sample, and are either detailed paraphrases or verbatim transcripts. For our application, we need a source that provides a comparable real-time, consistent, and detailed record of events in the cryptocurrency sector.

Our primary source is the Reuters News Archive. We restrict our search to English-language articles published via Reuters' real-time newswire that concern the cryptocurrency sector. The Reuters archive offers a continuous, timestamped, and verifiable record of events, free from the retrospective selection or editing biases that can arise in later summaries or analyses.

Reuters articles satisfy the requirements for a reliable narrative source along several dimensions. First, they are *contemporaneous*: articles are compiled in real time by professional journalists and include the precise date and time of release, direct quotations from relevant actors (e.g., government officials, regulatory bodies, corporate representatives), and a factual description of events. Second, Reuters' coverage is *consistent over time*: the editorial process and reporting style are standardized, and the crypto-sector newswire has been active throughout our sample period. Third, the source is *detailed and accurate*: Reuters operates under a rigorous editorial process that yields high factual accuracy, and has broad global coverage, reporting on developments from multiple jurisdictions. This allows us to capture a diverse range of regulatory, technological, and institutional news affecting the cryptocurrency market. Finally, Reuters prioritizes timeliness, meaning that news is often reported within minutes or hours of occurrence, ensuring that our event dates reflect the earliest public disclosure of new information.

One difference from the Romer and Romer setting is worth noting. The FOMC tran-

scripts were, for much of their sample, confidential—providing an unusually candid record of policymaker deliberations. Our source, by contrast, is public at the time of publication. However, this difference is a feature rather than a limitation in our context: we are not trying to ascertain the private motivations of policymakers, but rather to identify the arrival of *public* information about the cryptocurrency sector. What matters for our purposes is that the source accurately reflects what market participants could have known at each point in time—a criterion that a real-time newswire is ideally suited to satisfy.

Our sample covers January 2017 to December 2024. The starting date reflects the point at which cryptocurrency markets—and Bitcoin in particular—reached a level of liquidity and market structure comparable to other traded financial assets. By 2017, the market had developed several key features of a liquid asset class: the presence of multiple large exchanges operating globally, active market-making and cross-exchange arbitrage, and daily trading volumes exceeding $1 billion. Around the same time, the introduction of regulated derivatives—most notably Bitcoin futures on the Chicago Mercantile Exchange (CME) and the Chicago Board Options Exchange (CBOE) in December 2017—further deepened market activity and contributed to changes in trading volume and volatility dynamics (Menchetti et al., 2021). These developments mark the transition from a niche and fragmented market to one capable of generating price movements with potential spillovers to traditional financial assets and the broader economy.

For this sample period, Reuters maintained continuous, real-time coverage of the cryptocurrency sector. Rather than constructing ad hoc keyword searches, we rely on the news classification embedded in the Reuters system via the query `CRYCUR AND RTRS AND LEN`, corresponding to the Cryptocurrency topic code, Reuters original reporting, and the English language, respectively. This query provides a systematic and comprehensive selection of real-time news articles relevant to the sector.

2.2 Pillar 2: A Clear Idea of What One Is Looking For

Romer and Romer (2023) argue that a researcher must have a clear and specific idea of what they are looking for in the narrative source before beginning to read. In our setting, we look for episodes in which new, unanticipated information about the cryptocurrency sector became public—information that is plausibly exogenous to traditional macroeconomic and financial conditions. More precisely, we seek news events whose content is specific to the cryptocurrency market (e.g., regulatory actions targeting crypto, institutional adoption decisions, exchange failures, protocol changes) and whose timing is not mechanically linked to broader economic developments.

For each article, we assessed whether it contained significant, unanticipated information about the crypto sector. When ambiguity arose, the default was to exclude the article. To operationalize our search, we systematically evaluated every article against three exclusion filters:

1. **Insignificant developments:** We excluded news unlikely to alter aggregate cryptocurrency valuations, adoption expectations, or market functioning. Under this umbrella, we applied two standing exclusions: (a) news concerning central bank digital currencies (CBDCs) and stablecoins, which pertain to a distinct market segment and do not directly bear on the valuation of decentralized cryptocurrencies such as Bitcoin; and (b) announcements of new exchange launches, which consistently lack material price impact.
2. **Anticipated or non-exogenous events:** We excluded developments where the timing and content lacked novelty or were mechanically linked to prior events (e.g., scheduled protocol upgrades, pre-announced regulatory consultations, or items with credible advance signaling).
3. **Contaminated events:** We excluded episodes in which the cryptocurrency-related news was announced jointly with other major macroeconomic or policy measures, creating an inseparability of effects (e.g., crypto tax rules embedded within a national budget announcement).

Crucially, our assessment of an event's significance was based strictly on the text of the Reuters articles. We did not reverse-engineer the selection of events by first identifying days with large statistical price movements and then searching for the corresponding news. Instead, we relied solely on the information presented in the reporting. Naturally, if an article explicitly noted that a development coincided with a major market move (e.g., a 10% price drop), we took that text into account when judging the event's significance. However, we strictly refrained from matching candidate events to external, high-frequency price data at this stage; that systematic quantitative validation is the explicit purpose of Stage 2.

2.3 Pillar 3: Approaching the Source Dispassionately and Consistently

Romer and Romer (2023) stress that "to do the narrative analysis rigorously, one needs to approach the qualitative source dispassionately and consistently." They warn against the temptation to see what one wants to see in the narrative source, and recommend two safeguards: having multiple readers for the same document, and reading the entire source from beginning to end rather than focusing on selected periods.

We implement both safeguards. First, we rely exclusively on the contemporaneous narrative record, rather than retrospective analyses, opinion pieces, or market commentaries. The aim is to capture market-relevant surprises as they became public knowledge, not interpretations formed after the fact. To this end, we exclude Reuters "Explainer" articles, editorial columns, and speculative forecasts unless they introduce genuinely new, factual information.

Second, and critically, we read the *entire* corpus of retrieved Reuters cryptocurrency articles in full. This is a central feature of the narrative approach: it requires comprehensive reading of the source material rather than the targeted extraction of events based on ex post market outcomes. By reading chronologically without referencing external trading data, we mitigate the risk of selection bias arising from hindsight—the same risk Romer and Romer guard against by reading all FOMC transcripts rather than just focusing on meetings where they expect to find shocks. It ensures that events are identified based purely on the information set available to market participants at the time.

To operationalize this reading process, we divided the corpus into contiguous time segments and assigned each segment to a researcher. The division was structured to mix different years and market conditions across readers, so that all members of the research team developed familiarity with the full range of events and reporting styles present in the sample. For a subset of months, both researchers independently read and classified the same articles, enabling us to calibrate the application of the inclusion and exclusion criteria and to verify inter-rater consistency. This dual-reading procedure mirrors the Romer and Romer recommendation of "comparing classifications with another reader" to guard against individual bias.

2.4 Pillar 4: Documenting the Evidence

Romer and Romer (2023) emphasize that "writing detailed descriptions of our analysis of the narrative evidence is a crucial part of the work," noting that the documentation process clarifies thinking, sometimes changes classifications, and makes it easier for subsequent researchers to check the work. In this spirit, every candidate event in our dataset is accompanied by a brief description (`BriefDescription`), the original Reuters headline, and—for validated events—detailed annotations from both the human and LLM validators. The full event-by-event documentation is available at `[URL]`.

2.5 Final list of events

This process yielded a total of 298 candidate events spanning the sample period from January 2017 to December 2024. The events were then classified into mutually exclusive categories reflecting the primary nature of the news. Table 1 summarizes the classification scheme.

Table 1 Event Classification by Primary Nature

Category	Description
Regulatory and legal actions	New rules, enforcement measures, bans, sanctions, guidance issued by governmental or regulatory bodies
Institutional adoption or rejection	Payment integration, custody offerings, delistings, corporate treasury decisions
Sentiment and public discourse	High-profile endorsements or criticisms, media bans, public statements by influential figures
Other	Market structure changes, technological developments, exchange incidents, and events not fitting the above categories

Note. Each event is assigned to exactly one category. The corresponding binary indicators in the dataset are `Dregu`, `Dadop`, `Dsent`, and `Dothe`.

In the standard application of the narrative approach—for example, to monetary policy—the classification step is the endpoint: having read the FOMC transcripts and classified each meeting, the researcher has a shock series. In our setting, the open-ended nature of the event discovery problem means that Stage 1 necessarily casts a wider net. Some of the 298 candidate events may turn out to have had no discernible market impact, or to have been confounded by contemporaneous news. This is where our second identification strategy—high-frequency validation—enters. By confronting each candidate event with intraday Bitcoin price data in a tight window around the event timestamp, Stage 2 disciplines the narrative classification with market-based evidence, retaining only those events for which the narrative and the price response are mutually consistent.

3 Stage 2: Event Validation

The 298 candidate events identified in Stage 1 were screened on the basis of the contemporaneous narrative alone, without reference to price data. In Stage 2, each candidate event was

matched to intraday Bitcoin prices and subjected to a validation procedure that examined the joint behavior of the news narrative and price movements in a window spanning 12 hours before and 12 hours after the event timestamp.

The purpose of this second stage was to determine whether each candidate event was accompanied by a price reaction plausibly attributable to the news, or whether the event should be discarded because the price evidence was (i) absent, (ii) inconsistent, or (iii) confounded. This step has no direct analogue in Romer and Romer (2023), where the narrative classification is the final word. Two features of our setting make this additional validation both feasible and necessary. First, the availability of high-frequency Bitcoin price data—traded around the clock, globally—allows us to confront each candidate event with the market's contemporaneous reaction in a tight intraday window. Second, and more fundamentally, the open-ended nature of the event discovery problem described above means that Stage 1 necessarily casts a wider net than a classification exercise applied to a predefined set of meetings. Despite our best efforts to read the news carefully and consistently, some candidate events that appeared significant on the basis of the narrative alone may turn out to have had no discernible price impact (`No Move`), may have coincided with confounding news that makes attribution impossible (`Polluted`), or may have produced price movements inconsistent with the narrative (`Mismatch`). The price-based validation in Stage 2 disciplines the event selection by retaining only those events for which both the narrative and the price evidence point in the same direction.

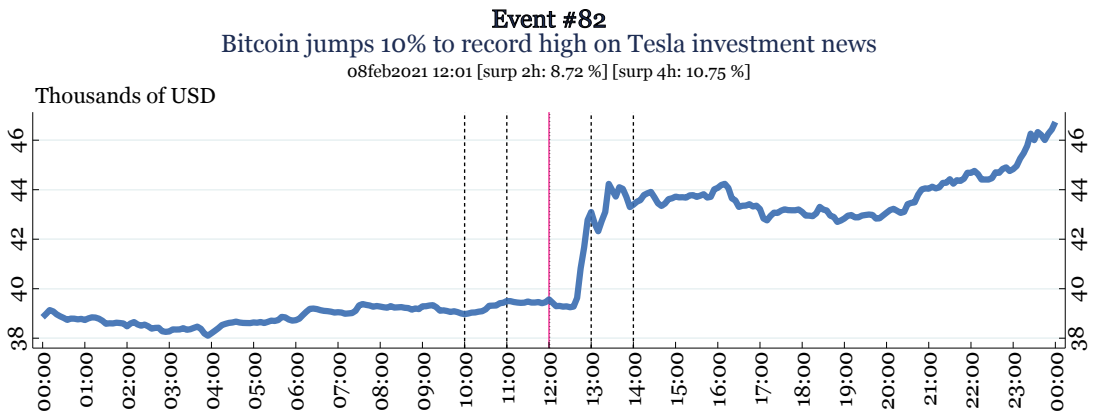
We implemented this validation through two independent approaches—one conducted manually by human researchers, the other conducted with the assistance of a large language model—each applying its own classification scheme. The dual-validation design is motivated by Romer and Romer's (2023) discussion of whether narrative research can be delegated to computers. They note that "for now, the main way to use the narrative approach is going to be the old-fashioned one—scholars sitting at their desks (or in our case, our dining room table) doing a lot of careful reading." But they also anticipate that delegation to computers may become feasible "at some point, in some cases, if it is done very carefully." Our LLM-assisted validation is designed as a test of exactly this proposition. By running both a traditional human validation and a structured LLM validation on the same set of events, we can assess the degree to which the two approaches agree, and whether the LLM can serve as an effective, reproducible second reader.

Both approaches are similar in spirit: they assess the alignment between the news narrative and the observed price response, and both result in a subset of validated events (those retained) and a set of discarded events (those dropped). The approaches differ in their implementation and in the granularity of their classification categories.

3.1 Human Validation

In the manual approach, each researcher re-read the original Reuters article alongside a standardized intraday Bitcoin price chart for the ±12-hour window around the event. Figure 1 provides an example of the chart used in this process.

Figure 1 Example of Standardized Event Chart Used in Validation Process



Note. Each chart displays a 5-minute-frequency Bitcoin price series spanning ±12 hours around the event timestamp (T_0), along with the news headline, date, time, and the 2-hour and 4-hour log returns.

Each event was assigned to one of the following categories:

(a) `Matched`: The news narrative is consistent with the observed price movement in terms of both direction and timing. The event is retained in the validated dataset.

(b) `Mismatch`: The observed price movement is in the opposite direction to what the news narrative would imply. The event is dropped.

(c) `No Move`: Despite the initial assessment in Stage 1 that the news was significant, the intraday price data show no discernible market reaction. The event is dropped.

(d) `Polluted`: Another major news event occurred within the same intraday window, making it impossible to attribute the observed price movement to the candidate event alone. The event is dropped.

Only events classified as `Matched` were retained in the manually validated dataset. In some cases, the news narrative was consistent with a price movement observed within the 24-hour horizon, but the movement occurred at a different time than the Reuters article timestamp. For these events, we conducted additional searches across other news sources (news agencies and financial websites) to identify whether the information had been released earlier. If a more precise timestamp was found, the event was retained as `Matched` with the adjusted timing (flagged via the `CommentHuman` variable; see Table 2); if not, the event was dropped. For all events with adjusted timing, the validated event timestamp (T_0) was

updated to reflect the earliest identified publication time, sourced from either an alternative news agency (`FirstAgencyGMT`) or a financial website (`FirstWebGMT`).

3.2 LLM-Assisted Validation

To complement the human validation, we developed a structured set of instructions and administered them to a large language model (GPT-5.4) to perform a systematic, highly structured validation of the same 298 candidate events. For each event, the model was provided with the same standardized PDF chart shown in Figure 1, containing the news headline, the date and GMT timestamp of the event (T_0), a 5-minute-frequency Bitcoin price chart spanning ± 12 hours around T_0, and the 2-hour and 4-hour log returns computed around the event.[3] The model was then instructed to:

(a) Parse the event metadata and return magnitudes from the PDF.

(b) Conduct a fresh set of web searches to retrieve contemporaneous news coverage from at least six sources (three crypto-focused and three general financial outlets). The purpose of this search was to verify the news, check for explicit price reporting that might contradict the data, and identify potential confounding events.

(c) Apply a four-step classification decision tree, evaluated in strict sequential order:

  (i) `No Move`: both the 2-hour and 4-hour absolute log returns are below 0.1

  (ii) `Mismatch`: contemporaneous articles explicitly report Bitcoin moving in one direction (e.g., "BTC surges"), but both observed return windows show a move in the opposite direction.

  (iii) `Polluted`: a specific, named confounding event identified through the web search (crypto-specific or macroeconomic) clearly explains the observed price movement.

  (iv) `Matched`: none of the above conditions are met (the residual category).

(d) For `Matched` events, assign a certainty rating on a 1–4 scale reflecting the strength of the plausible link between the news and the price movement. This scale ranges from 4 (price move is clearly consistent with the news interpretation; no alternative explanation) down to 1 (price move is opposite to the intuitive news interpretation, but no alternative explanation or confounding event was identified).

The complete set of instructions supplied to the model is reproduced in Appendix 6.3. A structured summary of the classification output for each event is available at `[URL]`.

As in the manual approach, only events classified as `Matched` were retained in the LLM-validated dataset. Events classified as `No Move`, `Mismatch`, or `Polluted` were dropped,

[3] To ensure the model evaluated the most accurate timeline, it was provided with the precise event timings (T_0) established during the human review process.

mirroring the logic of the manual validation. Notably, our definition of `Mismatch` for the LLM was deliberately narrow: it required an explicit contradiction between contemporaneous reporting and observed returns. Events with counter-intuitive price reactions (e.g., bad news followed by a price increase) that lacked explicit reporting contradictions or confounders were retained as `Matched` but assigned the lowest certainty score (1).

Each event was analyzed independently, with fresh web searches conducted for every event and no information carried over from the analysis of preceding events. The model's web searches retrieved contemporaneous source material—that is, articles published at or near the time of the original event—although the retrieval itself was conducted ex post during the construction of the dataset in 2025. This distinction is important: the *sources* consulted are contemporaneous, but the *access* to those sources occurred after the fact, not at the time of the original events.

3.3 Relationship Between the Two Approaches

The manual and LLM-assisted validation procedures are similar in spirit: both assess whether a candidate event's news narrative aligns with the observed intraday price behavior, and both result in a binary retain/discard decision. The key differences are operational rather than conceptual. The manual approach relies on direct human reading of the Reuters article and visual inspection of the price chart, producing a four-category classification. The LLM-assisted approach supplements the original Reuters article with a broader set of web-sourced contemporaneous coverage and applies a formalized, pre-specified decision tree with an additional certainty dimension for retained events.

By design, the two approaches were conducted independently. The result is two partially overlapping subsets of validated events drawn from the same pool of 298 candidates. This dual-validation structure serves as a robustness device: it allows us to examine whether the empirical results are sensitive to the choice of validation method and to characterize the events on which the two approaches agree or disagree. Summary statistics and a comparison of the two validated subsets are presented in Sections **??** and 6.

Our methodology is deeply indebted to the narrative tradition developed by Friedman and Schwartz (1963) and refined over decades by Romer and Romer. The core architecture of our approach—a reliable contemporaneous source, clear inclusion criteria specified ex ante, dispassionate and comprehensive reading, and detailed documentation—follows their template closely. Where we depart is in extending the approach to a new domain (cryptocurrency markets rather than monetary policy) and, more significantly, in introducing an LLM as a structured second reader.

This extension addresses a practical limitation that Romer and Romer (2023) themselves identify: the narrative approach is time consuming, and the question of reproducibility is paramount. Human readers, however careful, inevitably bring some subjectivity to the classification task. By running an LLM through the same material with a fixed, pre-specified set of instructions, we create an additional classification that is *fully reproducible*—any researcher with access to the same model, instructions, and inputs can replicate the LLM's output.[4] At the same time, we do not claim that the LLM replaces human judgment. Rather, the two approaches are complementary: the human reader brings contextual understanding and the ability to handle subtle or ambiguous cases, while the LLM brings consistency, scalability, and the ability to conduct systematic web searches across dozens of contemporaneous sources for each event. Together, they produce a richer and more robust classification than either could achieve alone.

Romer and Romer (2023) note that even for straightforward classification tasks, the correlation between computer algorithms and expert human readers has historically been modest—**?** highlights the challenges of extracting economic signals from narrative sources in the context of foreign exchange interventions, while Shapiro and Wilson (2022) report a correlation of about 0.4 between text-based measures and human assessments of Federal Reserve policymaker concerns. Our results, documented in Section 5, suggest that a carefully prompted LLM can achieve substantially higher agreement with human experts, at least for the type of narrative-plus-price validation task we study here. We view this as encouraging evidence that the frontier of what can be "carefully" delegated to computers has advanced meaningfully with the advent of large language models.

4 Final Dataset and Variable Definitions

The final dataset contains 298 events (rows) and 28 variables (columns). Columns A–M (variables 1–13) record the event identification, manual validation, and event-type classification produced during Stage 1 and the manual validation. Columns N–AB (variables 14–28) record the final event timing—incorporating any adjustments identified during the manual validation—and the full output of the LLM-assisted validation. Table 2 provides definitions for all variables. Missing values are denoted by "`.`" throughout.

[4]Subject to the usual caveats about model versioning and stochastic decoding. In practice, we fix the model version and use deterministic settings where available.

Table 2 Variable definitions for the event-level dataset

#	Variable	Description
Panel A: Event identification and manual validation (columns A–M)		
1	`Event_Number`	Sequential event identifier (1–298).
2	`ReutersGMT`	Original timestamp of the Reuters article (GMT).
3	`FirstAgencyGMT`	Earliest identified publication time from an alternative news agency (GMT). Used when the Reuters article was not the first to report the event. Equals `ReutersGMT` if no earlier agency source was found.
4	`FirstWebGMT`	Earliest identified publication time from a financial website (GMT). Populated only when an earlier web source was found during the adjusted-timing search; "`.`" otherwise.
5	`WebLink`	URL of the earliest web source identified, if applicable; "`.`" otherwise.
6	`PermaLink`	Archived (permanent) link to the earliest web source, if applicable; "`.`" otherwise.
7	`JudgeHuman`	Manual validation classification: `Matched`, `Mismatch`, `No Move`, or `Polluted`.
8	`CommentHuman`	Annotation flag. Set to "Alternative time" when the event was classified as `Matched` using an adjusted timestamp identified through additional source searches; "`.`" otherwise.
9	`Dadop`	Binary indicator equal to 1 if the event pertains to institutional adoption or rejection; 0 otherwise.
10	`Dregu`	Binary indicator equal to 1 if the event pertains to regulatory or legal actions; 0 otherwise.
11	`Dsent`	Binary indicator equal to 1 if the event pertains to sentiment and public discourse; 0 otherwise.
12	`Dothe`	Binary indicator equal to 1 if the event does not fall into the above categories (market structure, technological, or other); 0 otherwise.
13	`BriefDescription`	Short free-text description of the event.
Panel B: Timing and price data (columns N–Q)		

Continued on next page

Table 2 Variable definitions for the event-level dataset (*continued*)

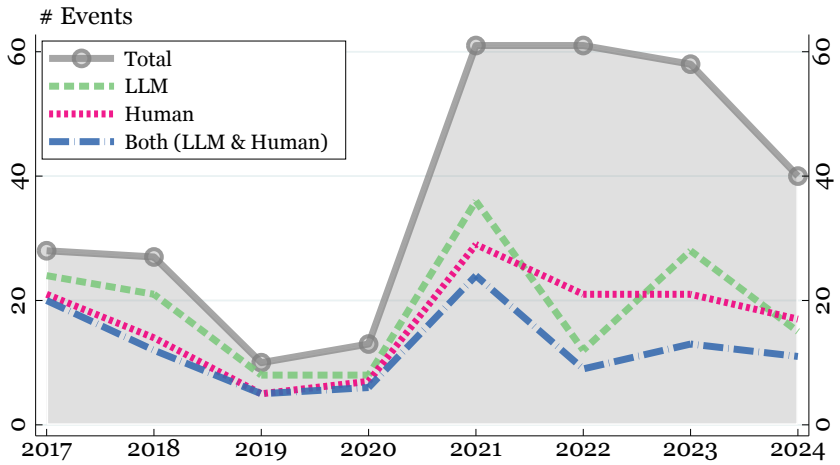
#	Variable	Description
14	`AnnouncementGMT`	Final event timestamp (T_0) used in the analysis (GMT). Equals `FirstWebGMT` if available; otherwise equals `FirstAgencyGMT`. Incorporates any timing adjustments from the manual validation.
15	`Date_GMT`	Calendar date of the event.
16	`Time_T0_GMT`	Time component of T_0 (HH:MM, GMT).
17	`Year`	Calendar year of the event.
Panel C: LLM-assisted validation (columns R–AB)		
18	`JudgeAI`	LLM validation classification: `Matched`, `Mismatch`, `No Move`, or `Polluted`.
19	`Certainty`	Certainty rating for `Matched` events (1–4); `NA` for all other classifications. 4 = very high confidence, no confounders; 1 = price moves opposite to narrative with no alternative explanation.
20	`Return_2h`	2-hour log return around T_0 (percentage).
21	`Return_4h`	4-hour log return around T_0 (percentage).
22	`Window`	Indicates which return window(s) align with the news narrative for `Matched` events: `2h`, `4h`, `Both`, or `NA`.
23	`News_Headline`	News headline as recorded on the event PDF.
24	`News_Source`	Primary source(s) of the news as identified by the LLM.
25	`Timing_Assessment`	LLM assessment of when the price move began relative to T_0.
26	`Confounds`	LLM assessment of whether confounding events were identified within the event window.
27	`Narrative_Fit`	LLM one-sentence assessment of the alignment between the news narrative and the observed price movement.
28	`Sources`	List of sources consulted by the LLM during the systematic web search.

5 Event Classification and Agreement

This section assesses the degree of agreement between the human and LLM validators—providing an empirical answer to the question raised by Romer and Romer (2023) about the feasibility of delegating narrative analysis to computers. We document the classification outcomes, examine the confusion matrix between the two approaches, and characterize the composition of validated events by type.

Figure 2 displays the number of candidate events per year together with the subsets retained by the human and LLM validators, and their intersection. The total number of candidate events trends upward over the sample, reflecting the growing volume of cryptocurrency news coverage on Reuters. The human- and LLM-matched series track each other closely and are substantially smaller than the pool of 298 candidates, as expected given the validation described in Section 3. The intersection of the two—events retained by both validators—forms the most restrictive classification and is used as the baseline in our empirical analysis.

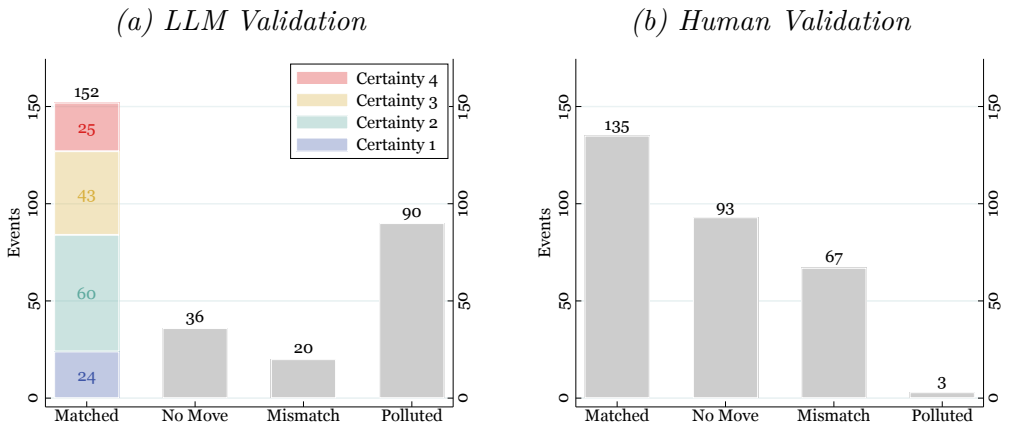
Figure 2 Events by Year: Total, LLM-Matched, and Human-Matched



Note. Number of events per year. Grey area = total candidate events (298); green dashed line = LLM-matched events; pink line = human-matched events; blue dash-dot line = events matched by both human and LLM (human ∩ LLM).

Figure 3 breaks down the classification outcomes by category for each validator. For the LLM, Matched events are further stratified by certainty level (1 = lowest to 4 = highest). The distribution across categories is broadly comparable across the two validators: Matched events constitute the largest group, while No Move, Mismatch, and Polluted account for similar shares. Among LLM-matched events, the majority carry high certainty ratings (levels 3 and 4), lending credibility to the automated classification.

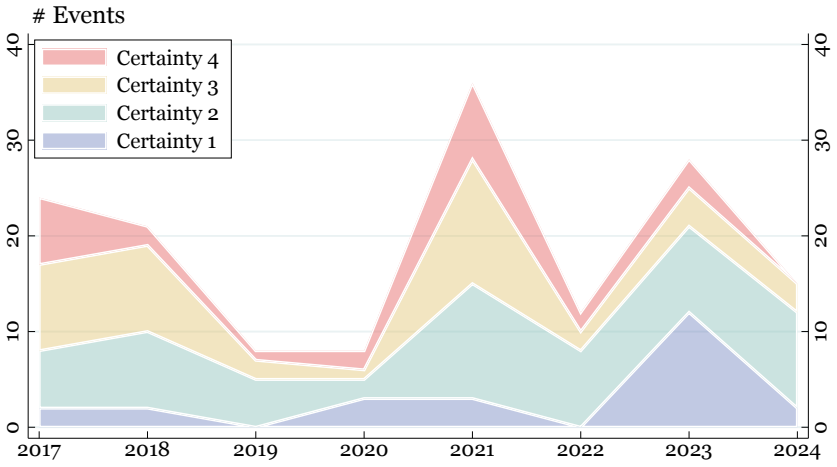
Figure 3 Event Classification by Category



Note. Number of events per classification category (Matched, No Move, Mismatch, Polluted) on the horizontal axis. Panel (a): LLM classification, with Matched bars further split by certainty level (1 = lowest to 4 = highest). Panel (b): Human classification.

Figure 4 shows how LLM-matched events and their certainty ratings evolve over time. High-certainty matches (levels 3–4) dominate throughout the sample, and the volume of matched events grows in line with the overall increase in crypto market activity. Lower-certainty matches are somewhat more prevalent in the early part of the sample, when the Reuters corpus was thinner and contemporaneous corroboration harder to find.

Figure 4 LLM-Matched Events by Certainty Level, Over Time



Note. Stacked area chart of LLM-matched events per year, split by certainty level (1 = lowest to 4 = highest).

Table 3 presents the *confusion matrix* of human and LLM validation outcomes, with rows corresponding to human judgments and columns to LLM judgments. The main diagonal captures agreement; off-diagonal mass is concentrated in adjacent categories, most commonly

when one validator classifies an event as `Matched` and the other as `No Move`. The raw percent agreement is 58.7% and Cohen's κ is 0.342—formally "fair" agreement, but economically meaningful given the complexity of the task and the historically modest concordance in related narrative exercises (Naef, 2024; Shapiro and Wilson, 2022).[5]

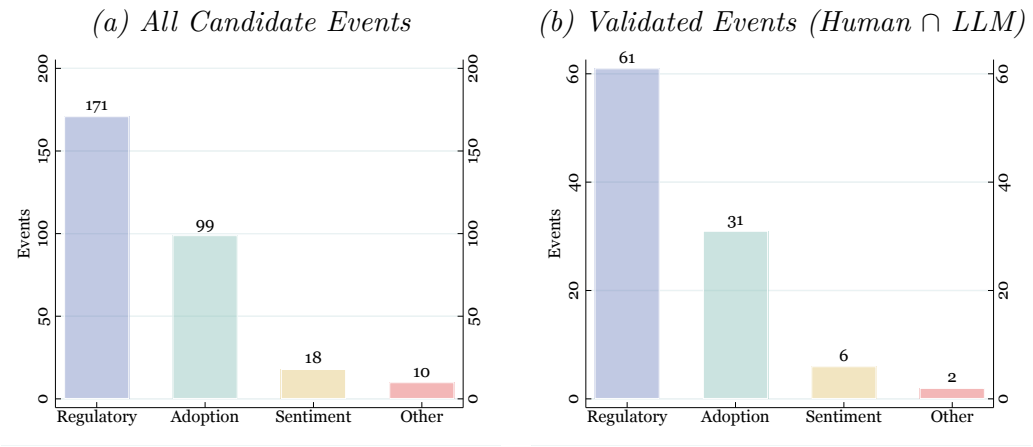
Table 3 Human vs. LLM Classification Confusion Matrix

	LLM			
Human	Matched	No Move	Mismatch	Polluted
Matched	100	3	2	30
No Move	31	27	5	30
Mismatch	19	6	13	29
Polluted	2	0	0	1

Note. Rows report manual classifications and columns report LLM classifications. Raw percent agreement = 47.3%; Cohen's $\kappa = 0.261$.

Figure 5 displays the composition of events by type. Panel (a) shows all 298 candidate events; panel (b) restricts attention to events retained by both the human and LLM validators (the intersection sample used as the baseline throughout the paper). Regulatory and legal actions constitute the largest category in both panels, followed by adoption, sentiment, and other events.

Figure 5 Event Composition by Type



Note. Number of events by type: Regulatory and legal actions, Institutional adoption or rejection, Sentiment and public discourse, and Other. Panel (a): all 298 candidate events. Panel (b): events retained by both human and LLM validators.

[5]Under the **?** benchmarks, $\kappa = 0.342$ falls in the "fair" range (0.21–0.40). In skewed multi-category tasks of this type, values in this range indicate that the LLM captures a significant portion of the human researcher's underlying classification logic.

6 Measuring Crypto Surprises & Stylized Facts

Having documented the classification agreement between the human and LLM validators, we now turn to the surprise series itself—its construction, its properties across validation methods, and its sensitivity to the choice of event window.

6.1 Definition of the Surprises

For each validated event, we measure the intraday Bitcoin price response in a 2-hour window centered on the event timestamp, constructing a series of *crypto surprises* that capture unanticipated price movements tightly around each event. This high-frequency measurement strategy, in the spirit of Gürkaynak et al. (2005), isolates rapid market adjustments triggered by the identified cryptocurrency-specific events while mitigating potential confounding effects from broader financial market fluctuations. The choice of a ±1-hour window balances the need to capture timely price adjustments against the risk of contamination from unrelated market movements and potential measurement errors in the timing of events.

We focus on the Bitcoin spot price, which serves as the benchmark for pricing in the cryptocurrency market and has the longest available history among crypto-traded instruments. Our price data come from Kraken, a US-based cryptocurrency exchange founded in 2011, with a long operating history and consistently high trading volumes.[6] Kraken provides minute-frequency data on opening and closing prices, high and low prices, trade volume, and trade count.

We construct the surprise for each event as the log difference of Bitcoin spot prices one hour after and one hour before the event:

$$s_{t,k} = \log\left(\bar{P}_{t,k+60'}\right) - \log\left(\bar{P}_{t,k-60'}\right), \tag{1}$$

where k denotes the event timestamp and t denotes the calendar day. Because the underlying data are sampled at 5-minute frequency and individual observations can occasionally be missing, we define $\bar{P}_{t,k+60'}$ as the average of the Bitcoin spot price at minutes $k+55'$, $k+60'$, and $k+65'$, and similarly $\bar{P}_{t,k-60'}$ as the average at minutes $k-65'$, $k-60'$, and $k-55'$. In other words, each endpoint price is measured as a local 15-minute average centered on the target minute rather than as a single print. This averaging smooths out microstructure noise, reduces sensitivity to an isolated missing quote, and ensures that the surprise is not

[6] As of 2024, Kraken processed on the order of \$600–700 billion in annual trading volume and held over \$40 billion in client assets, placing it among the major global cryptocurrency exchanges. Founded in 2011, it is also one of the longest-operating platforms in the sector.

driven by a single potentially stale price observation.[7]

Table 4 reports summary statistics for the resulting surprise measure. The table distinguishes between the raw all-events series and the subsets retained by the LLM validator, the human validator, and their intersection. The matched subsets display similar central tendencies and dispersion, which is consistent with the close alignment documented later in this section.

Table 4 Crypto Surprises: Summary Statistics

	All events	LLM (Matched)	Human (Matched)	Intersection
Mean	-0.06	-0.03	-0.07	-0.10
Median	0.01	0.17	0.15	0.29
Standard Dev.	2.14	2.81	3.05	3.42
Skewness	-1.33	-1.21	-1.01	-0.98
Kurtosis	33.71	22.41	18.35	15.66
Minimum	-19.52	-19.52	-19.52	-19.52
Maximum	15.06	15.06	15.06	15.06
Pos./Neg. ratio	1.04	1.24	1.18	1.27
Observations	298	152	135	100

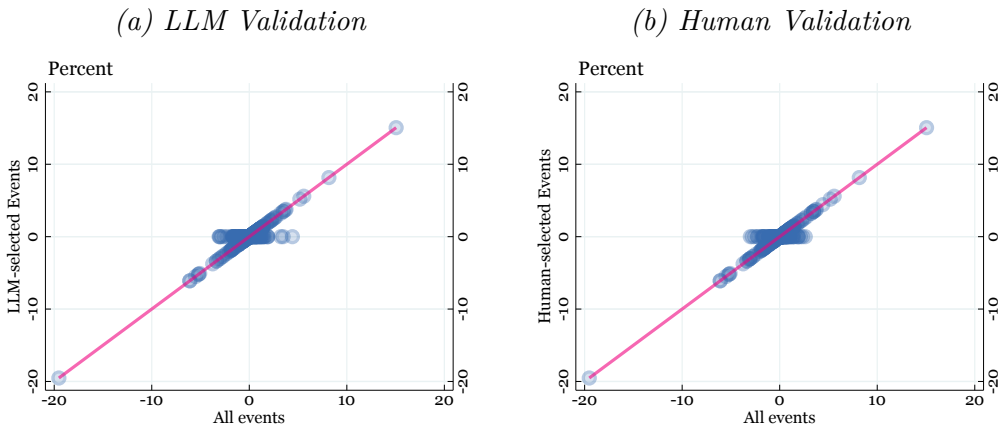
Note. Summary statistics for 2-hour crypto surprises by classification. Pos./Neg. ratio denotes the number of positive shocks divided by the number of negative shocks, excluding zero shocks. The sample covers the period between Jan 2017 and 31 Dec 2022.

6.2 Properties of the Surprise Series: All, Human, and LLM

Figure 6 plots the validated surprise measures against the "all-events" surprise constructed from all 298 candidates without any validation—panel (a) for the LLM classification and panel (b) for the human classification. Events discarded by the classifier appear as deviations from the 45-degree line, sitting flat along the zero line of the vertical axis. As expected, many of these discarded events also cluster near zero on the horizontal axis (representing `No Move` cases). However, some discarded events show large horizontal values but zero vertical values, representing cases where significant price volatility was present but ultimately classified as `Polluted` or a `Mismatch`.

[7] As a robustness check, we also construct alternative surprise measures using a 1-hour window (±30 minutes) and a 4-hour window (±2 hours). In both cases, we apply the same three-point averaging rule at the relevant window endpoints.

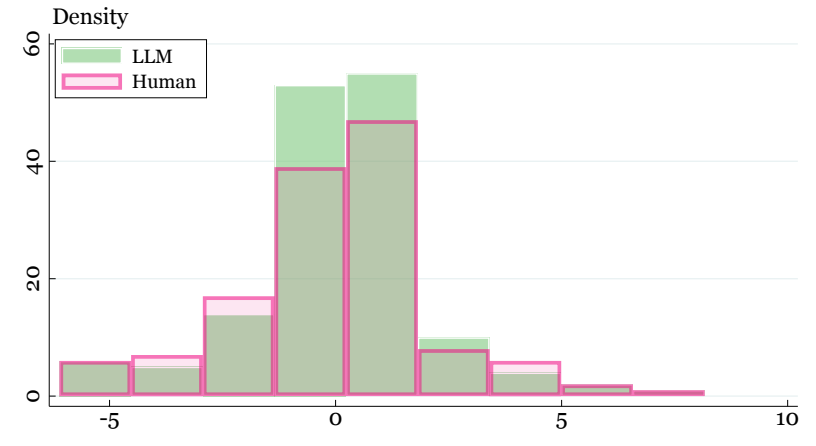
Figure 6 Validated vs. All-Events Surprise



Note. Scatter of validated surprises (vertical axis) against the unfiltered surprise based on all 298 candidate events (horizontal axis). Pink line = 45-degree reference. Events discarded by the validator appear near zero on the vertical axis. Panel (a): LLM validation. Panel (b): Human validation.

Having established that the validation step removes events carrying little price information, we now turn to comparing the human- and LLM-validated surprise series directly. The raw distributions feature two extreme outliers—one large positive surprise and one large negative surprise—both of which are retained by both the human and LLM validators. To ensure that the comparison is not dominated by these two observations on which there is concordance, Figure 7 overlays the trimmed distributions (1st–99th percentiles) of the LLM- and human-validated surprise series. The two histograms are strikingly similar, suggesting strong agreement on the economically relevant events.

Figure 7 Surprise Distribution: LLM vs. Human Validation

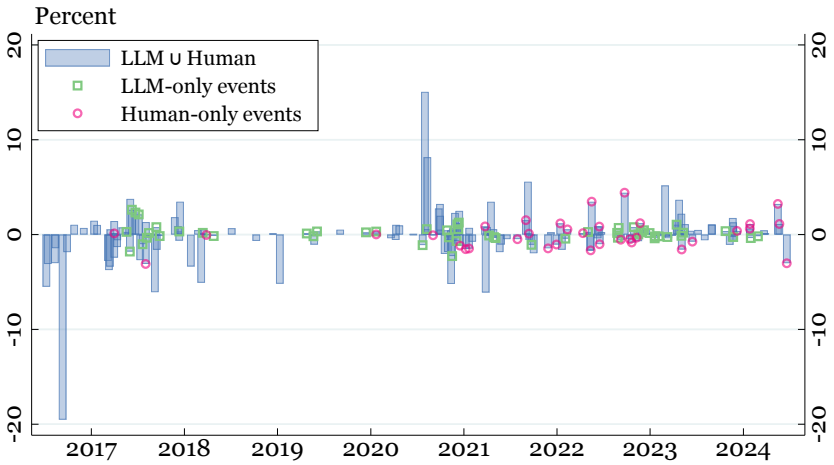


Note. Histogram of non-zero daily surprises for LLM-validated events (green) and human-validated events (pink), trimmed at the 1st–99th percentiles.

To visualize the rare cases of disagreement, Figure 8 displays the union of the two sur-

prise series (LLM ∪ human), marking days where only the LLM retained the event (green diamonds) or only the human retained it (pink circles). Disagreements between the two validators are relatively rare and tend to involve smaller surprises, while the large, economically impactful events are typically agreed upon by both the human and LLM classifiers. The intersection of the two—events retained by both the human and LLM validators—constitutes the most conservative measure and is used as the baseline throughout the paper.

Figure 8 Surprise Series: Union of LLM and Human (Disagreements Marked)



Note. Time series of the daily crypto surprise (log Bitcoin price change in the 2-hour event window, vertical axis) for the union of LLM- and human-validated events (LLM ∪ human, blue line). Green diamonds mark days where only the LLM retained the event; pink circles mark days where only the human retained it. Days with no marker indicate agreement between the two validators.

Taken together, the evidence in this section suggests that the human and LLM validators produce strikingly similar classifications. The large, economically meaningful events are identified by both approaches, and disagreements are concentrated among smaller or more ambiguous cases. This level of concordance is considerably higher than the modest correlations reported in prior comparisons of computer algorithms and human readers for related narrative tasks (Shapiro and Wilson, 2022), and supports the view that carefully prompted large language models represent a meaningful advance in the scalability and reproducibility of narrative identification.

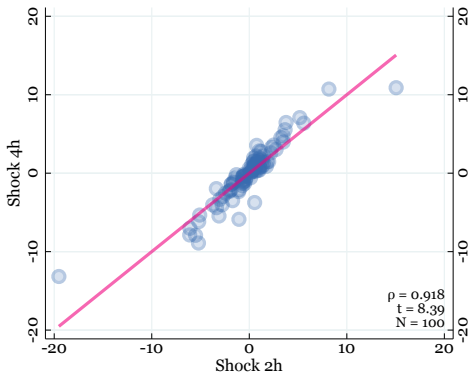
6.3 Properties of the Surprise Series: Window Length

A natural concern is whether the surprise series is sensitive to the choice of event window. To assess this, we compare the baseline 2-hour surprise (±1h around the event timestamp) with two alternatives: a narrower 1-hour window (±30 minutes) and a wider 4-hour window

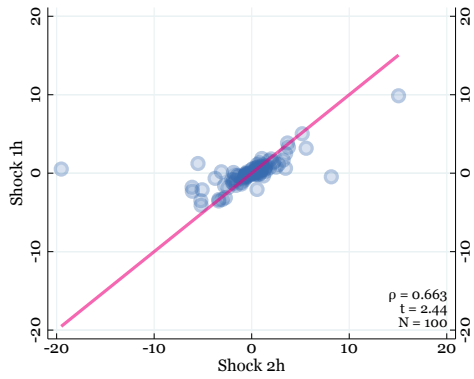
(±2 hours). Figure 9 plots the baseline surprise against each alternative for events retained by both the human and LLM validators.

Figure 9 Surprise Series Across Event Windows

(a) Baseline (2h) vs. Wide Window (4h)



(b) Baseline (2h) vs. Narrow Window (1h)



Note. Scatter of daily crypto surprises computed over the baseline 2-hour window (±1h) against alternative windows, for events retained by both human and LLM validators. Pink line = 45-degree reference. Panel (a): wide window (±2h). Panel (b): narrow window (±30min).

In both cases, the observations cluster tightly along the 45-degree line, indicating that the surprise series is largely invariant to the window width. The correlation between the 2-hour and 4-hour series exceeds 0.9. The correlation between the 2-hour and 1-hour series is somewhat lower, at around 0.67, but this is largely driven by a single observation that registers a large negative surprise in the 2-hour window yet zero in the 1-hour window. This pattern highlights the value of using a wider measurement window: in some cases, the price response to news unfolds gradually, so that a narrower window fails to capture the full adjustment. Overall, these results suggest that the identified events trigger price adjustments that do not reverse or amplify substantially over longer horizons, while confirming that the baseline 2-hour window strikes a reasonable balance between capturing the complete market response and limiting exposure to confounding developments.

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LLM Instructions

This section reproduces, verbatim, the complete set of instructions supplied to the language model (Claude, Sonnet 4, Anthropic) for the LLM-assisted validation described in Section 3.2. The instructions were provided to the model at the start of each conversation session and remained fixed throughout the project.

Crypto News Analysis Instructions

I have some instructions for a project on ‘‘crypto shocks’’. The project seeks to identify episodes (‘‘events’’) that affect crypto prices but that do not originate from the traditional financial sector, the real economy, or economic policies. As such, the price movements around these events can be thought as exogenous to macro-financial conditions.

Input

For each event, I will provide a PDF containing:

- A news headline regarding the crypto market
- The date of the news
- The time when the news was released (T_0, GMT 24h)
- A chart of BTC intraday moves (5-minute frequency, $\pm$12h around T_0)
- The BTC log-return in a 2h window ($\pm$1h around T_0) and 4h window ($\pm$2h around T_0)

Goal

Determine whether the news is plausibly linked to the BTC price move in the 4-hour window (i.e. $\pm$2h around T_0), and assess the degree of certainty of such assessment.

Step-by-Step Analysis Process

1. Parse PDF Information

Extract complete information from PDF: Event number; News headline; Date and time (T_0); 2h and 4h BTC log-returns (positive log-returns indicate price increases, negative indicate decreases). This is a crucial step, and information from parsed PDF is the basis for analysis that follows.

2. Systematic News Search

Based on the news headline and date from the PDF, conduct comprehensive news search covering crypto-related news from 12 hours before to 12 hours after T_0, plus major financial/macro news (data releases, policy actions, Fed announcements, etc.) for the same calendar day. Search minimum 3 crypto news sources (e.g., CoinDesk, CoinTelegraph, The Block) and 3 financial sources

(e.g., Reuters, Bloomberg, WSJ). The purpose is to identify confounding events and check for explicit BTC price reporting that might contradict actual returns in the 2h and 4h windows.

3. Apply Decision Tree Classification

- **Step 3a: Check for No Movement**
  - If absolute value of both 2h and 4h returns < 0.1% → No Move
- **Step 3b: Check for Explicit Price Contradictions**
  - If systematic search finds articles explicitly stating ''BTC rises/increased/up'' but both 2h and 4h returns are negative (< --0.1%), OR articles state ''BTC falls/decreased/down'' but both returns are positive (> +0.1%) → Mismatch
- **Step 3c: Check for Confounding Events**
  - If systematic search reveals alternative events (crypto-specific OR macro) that occurred within ±2h of T_0 and are explicitly linked to BTC price moves of similar or greater magnitude in contemporaneous reporting → Polluted
- **Step 3d: Confirm events**
  - Everything else → Matched

Categories examples

- No Move: 2h: +0.05%, 4h: --0.08%
- Mismatch: News says ''BTC surges on adoption news'' but 2h: --2.1%, 4h: --1.8%.
- Polluted: Event about exchange hack, but Fed announces rate cut 30 minutes before T_0 with explicit BTC price attribution
- Matched (Certainty 4): ''Major exchange adds BTC'' at 14:00, 2h: +3.2%, 4h: +2.8%, move starts 14:05

Certainty Scale

Applied only to Matched events:

- 4 (Very Much): price move clearly consistent with news interpretation; no credible alternative explanation.
- 3 (Good): price move consistent with news interpretation; some background volatility, no credible alternative explanation.
- 2 (Moderate): news is neutral/ambiguous, hard to link price move with news interpretation. Timing ambiguity or high volatility.
- 1 (Very Little): price move opposite to news interpretation; no credible alternative explanation.
- NA: Used for Polluted, Mismatch, and No Move categories.

Window Analysis

Track what happens in both 2h and 4h windows. Record under a field ‘‘Window’’ which window better aligns with news narrative:

- Both (if both returns have same sign and align with news)
- 2h (if only 2-hour return aligns with news)
- 4h (if only 4-hour return aligns with news)
- NA (for all non-matched categories and for matched with certainty 1)

Reporting Format

Compact Summary Line:

- Event number -- Date GMT -- Category | Certainty: {1/2/3/4/NA} | T_0: {HH:MM GMT} | Move: {2h return, 4h return}

Six Analysis Bullets:

- News: Short factual description of headline and source
- Market impact: Exact 2h/4h returns from PDF with brief interpretation
- Timing: Analysis of when move starts relative to T_0, based on 5-minute chart and returns
- Confounds: Note if other events of similar importance are present
- Narrative fit: One-sentence judgment linking news to price move using assigned category
- Sources: List of sources examined during systematic search

Dataset Building

After completing each event analysis, automatically create or update two artifacts:

CSV Dataset Artifact:

- Artifact ID: crypto_analysis_dataset
- Type: application/vnd.ant.code with language=‘‘csv’’
- Action: For first event, create new artifact. For subsequent events, update by adding new row to existing dataset
- Headers: Event_Number, Date_GMT, Time_T0_GMT, Category, Certainty, Return_2h, Return_4h, Window, News_Headline, News_Source, Timing_Assessment, Confounds_Present, Narrative_Fit_Summary, Sources_Searched

Markdown Documentation Artifact:

- Artifact ID: crypto_analysis_markdown
- Type: text/markdown
- Action: For first event, create new artifact. For subsequent events, update by appending new event section
- Content: Title (Event number + short heading based on news); Compact summary line; Six bullets

This step is mandatory and should be completed immediately after providing the analysis for each event, without requiring user prompts.

Practical info

- I will upload 5 pdfs at the time, and a maximum of 30 PDFs per chat. Please confirm that this approach will not lead to a deterioration of quality of your searches.
- Remember to run fresh, independent news searches for each event's specific date/time. No carryover bias between events. No memory should be used to make any assessment, only the info from the uploaded PDFs and the searches.